

LESSON 2-2 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Building a Histogram**

Maryland Standard 6.SP.B.4 • Learning target – I can group data into equal intervals and build a histogram from a frequency table.

REMEMBER

- A **histogram** groups data into **equal intervals**, like 0-9, 10-19, 20-29.
- The **frequency** is how many values fall in an interval. It sets the bar height.
- Intervals do not overlap, and the bars **touch**.

WORKED EXAMPLE

Minutes of practice: 4, 15, 22, 9, 18, 27, 13, 16, 21, 11.
Make a frequency table.

Step 1 0-9: 4, 9, so the frequency is 2.

Step 2 10-19: 15, 18, 13, 16, 11, so it is 5.

Step 3 20-29: 22, 27, 21, so it is 3.

Answer 2, 5, 3

WORD BANK

interval – a group of numbers, like 10-19

frequency – how many numbers are in a group

histogram – a graph with bars that touch

overlap – share the same number

PRACTICE

show your work • label units

- 1. Points Scored.** A player scored these points in 12 games: 8, 14, 21, 5, 17, 12, 25, 19, 11, 23, 16, 9. Complete the frequency table.

Step 1 Find the numbers from 0 to 9. Count them.

Step 2 Do the same for 10-19 and 20-29.

Step 3 Check: the counts add up to 12.

Points	Tally	Frequency
0-9		
10-19		
20-29		

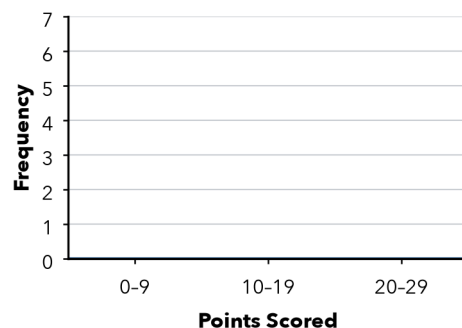
- 2.** Draw the histogram for Problem 1. Bars should touch.

Step 1 Use the counts from Problem 1 for the bar heights.

Step 2 Draw the bars so they touch.

a. Which interval has the tallest bar? _____

b. Games with 10 or more points: _____



- 3. Picnic Ages.** 34, 22, 41, 38, 27, 45, 31, 36, 29, 48 years. Find each frequency.

a. 20-29: _____

b. 30-39: _____

c. 40-49: _____

- 4. Missing Bar.** A histogram shows **24** students. Three bars are **5**, **9** and **3** tall. There is one more bar. How tall is it?

Step 1 Add the three bars you know.

Step 2 Subtract that sum from 24.

Answer _____ students

- 5. Find the Error.** Marcus used the intervals 0-9, 10-14, and 15-29. Explain his mistake and fix it.

Frame Marcus's intervals are not ____. Better intervals are ____, ____ and ____.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____

AT HOME Ask family members how many minutes they read today. Group the answers in intervals of 10.

LESSON 2-2 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Describing Histogram Shape**

Maryland Standard 6.SP.B.4 • Learning target – I can describe the shape of a histogram using its peak, spread, and balance.

REMEMBER

- **Peak:** the tallest bar. **Symmetric:** sides match.
- **Skewed right:** the tail stretches right. **Skewed left:** the tail stretches left.
- A histogram shows counts, not exact values.

WORKED EXAMPLE

Bars of height 2, 7, 4, 2, 1 over 0-9, 10-19, 20-29, 30-39, 40-49. Shape?

Step 1 The peak is 10-19, near the left.

Step 2 The bars shrink to the right: a tail on the right.

Answer Skewed right

WORD BANK

peak – the tallest bar

symmetric – both sides match

skewed right – long tail on the right

skewed left – long tail on the left

PRACTICE

show your work • label units

- 1. Test Scores.** The histogram shows the scores on a class test.

Step 1 Add all the bar heights for a.

Step 2 The tallest bar is the peak.

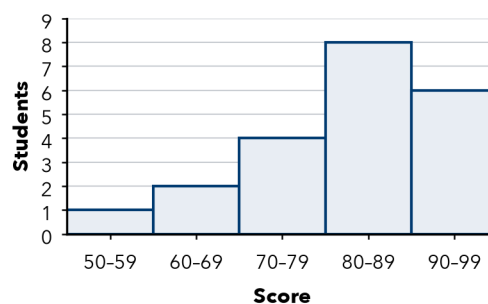
Step 3 Look for the long tail to name the shape.

a. Number of students: _____

b. Peak interval: _____

c. Shape: _____

d. Scores of 80 or more: _____



- 2.** Bar heights are listed from left to right. Name each shape.

Step 1 Find the biggest number in each list.

Step 2 Look at which side has the small numbers: that is the tail.

a. 1, 4, 8, 4, 1 _____ **b.** 9, 6, 3, 2, 1 _____ **c.** 1, 2, 3, 6, 10 _____

d. 3, 5, 9, 5, 3 _____ **e.** 12, 8, 5, 3, 2 _____ **f.** 2, 3, 5, 8, 12 _____

- 3. Race Times.** 30 runners. Bars of **3, 8, 11, 6, 2** show times of 20-24, 25-29, 30-34, 35-39, 40-44 minutes.

a. How many runners finished in less than 30 minutes? _____

b. How many finished in 35 minutes or more? _____

4. **Stretch.** Write 5 bar heights, from left to right, for a histogram of **20** values that is **skewed left**.

Bar heights _____

5. **Explain.** Ramon says the Test Scores histogram shows a score of 85. Is he right? Explain.

Frame Ramon is ____ because a histogram only shows ____.

TEST PRACTICE

choose the best answer

6. Bar heights are listed from left to right. Which histogram is skewed right?

A 2, 5, 9, 5, 2

B 10, 7, 4, 2, 1

C 1, 2, 4, 7, 10

D 5, 5, 5, 5, 5

7. Bars of height 4, 7, and 5 sit over 0-9, 10-19, and 20-29. How many values are 10 or more?

A 7

B 11

C 12

D 16

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #_____